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Ultra-Compact Low-Loss 3D Glass Waveguide Chip Using Femtosecond Laser Writing Technique

Fengrui Yu¹  | Lin Ma¹  | Mingjing Xu¹ | Xiaoke Chen¹  | Jinhua Wu² | Zuyuan He¹

¹State Key Laboratory of Photonics and Communications, Shanghai Jiao Tong University, Shanghai, China | ²Shanghai Optoweave Technology Co. Ltd., Shanghai, China

Correspondence: Lin Ma (ma.lin@sjtu.edu.cn)

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ABSTRACT

Interfacing multicore fiber (MCF) with high-speed optical transceivers serves as a promising solution for large-scale artificial intelligence (AI) data centers. Fan-in/fan-out (FI/FO) devices are critical components for interconnection. However, the stringent space and loss budgets in optical transceivers make the realization of integrable FI/FO devices a significant challenge. In this work, we demonstrate an ultra-compact low-loss FI/FO device for 4-core fiber using femtosecond laser direct writing (FLDW). The device features a footprint as small as $2.5 \times 2.0 \text{ mm}^2$ and an insertion loss of 0.54 dB at the wavelength of 1310 nm, serving as a critical interface between MCFs and high-speed optical transceivers. Through multi-scan and overwriting methods, we achieve low-loss waveguides with a propagation loss of 0.13 dB/cm and a coupling loss of 0.06 dB/facet. To mitigate bending loss, we propose a trench-assisted and offset-assisted cosine-shaped waveguide (TOA-cos). In this architecture, auxiliary trenches effectively suppress radiation loss, while offset-assisted cosine curves minimize mode-mismatch loss. Moreover, the TOA-cos model exhibits compatibility with existing FLDW bending-loss reduction methods, such as integrated microcracks and composite waveguides, making it a practical solution for chip-level three-dimensional optical interconnects.

1 | Introduction

The rapid expansion of data centers, cloud computing, and artificial intelligence (AI) is driving the demand for optical transceivers with higher transmission capacity [1–3]. For next-generation 1.6T optical transceivers, conventional scaling methods face fundamental limitations: increasing fiber count elevates system complexity and reduces reliability [4], while coarse wavelength division multiplexing (CWDM) suffers from the restricted transmission distance due to severe chromatic dispersion at 400 Gbit/s per fiber [5]. These challenges render existing solutions impractical for modern large-scale AI data centers. Multicore fibers (MCFs), with their superior spatial multiplexing capability, have emerged as a promising alternative [6]. By integrating multiple channels within the same cladding diameter as single-mode fibers (SMFs), MCFs reduce the overall system footprint and support extended

transmission distance at high baud rates via single-wavelength operation [5]. To interface MCFs with photonic integrated circuits inside optical transceivers, fan-in/fan-out (FI/FO) devices are indispensable. A critical design challenge for FI/FO devices, similar to the thin-film-filter-based WDM blocks or glass interposers [7, 8], is the stringent space budget inside optical transceivers, which confines their length to only a few millimeters. Consequently, realizing ultra-compact, low-loss, and integrable FI/FO devices has emerged as a critical challenge. Given the intrinsic three-dimensional geometry of FI/FO devices, femtosecond laser direct writing (FLDW) has therefore become a pivotal enabling technique for fabricating the required compact three-dimensional photonic interconnect chip (3D-PIC) [9].

Femtosecond lasers, characterized by their ultrashort pulse duration and high peak energy density, have been widely utilized

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in waveguide inscription. In contrast to conventional photolithography technique, FLDW is a rapid prototyping technique, enabling maskless fabrication in transparent bulk glasses, such as fused silica, boro-aluminosilicate glass, chalcogenide glass, and phosphate glass [10–13]. FLDW utilizes a precise and flexible focal point positioning mechanism to alter the refractive index of glasses, enabling the fabrication of 3D waveguides for integrated photonic devices [14–16]. Since these structures are embedded within the bulk glass substrates, the resulting waveguides exhibit high physicochemical stability and reliability, rendering them highly suitable for more complex optical communication environments [17]. Benefiting from simple fabrication procedures, truly 3D processing capability, and ultra-high stability, FLDW has facilitated significant advancements in various applications, including optical interconnects [18, 19], topological photonics [20, 21], quantum optics [22, 23], and astrophotonics [24, 25].

The push for ultra-compact, low-loss, and integrable FI/FO devices demands superior bending performance from FLDW waveguides [26]. The bending loss in curved waveguides includes scattering loss, radiation loss, and mode mismatch loss [27]. Scattering loss arises from the sidewall roughness [28]. Radiation loss is dependent on the bending radius of curved waveguides. Mode mismatch loss occurs at the interfaces between two waveguides with different curvature. Abrupt curvature changes are the primary cause of bending loss at relatively small radii, where radiation loss remains tolerable, as they induce significant mode mismatch [27, 29]. To date, extensive researches have been conducted to reduce the bending loss in FLDW waveguides, such as bend-loss-suppression walls, integrated microcracks, composite waveguides and multiple scanning [30–36]. Although Ross-Adams et al. recently demonstrated a record bend radius (1.0 dB/cm at 4.0 mm, 1550 nm) using a hybrid thermal-athermal waveguide design [33], this approach is hindered by the high insertion losses at the thermal-athermal interfaces. Additionally, these prior approaches primarily focus on increasing the refractive index contrast between FLDW waveguides and substrates to mitigate radiation loss. Mode mismatch loss has not been systematically minimized, which offers a promising alternative pathway for further reducing bending loss in FLDW waveguides.

We demonstrated an ultra-compact low-loss FI/FO device using FLDW, serving as a key interface interconnecting MCFs with high-speed optical transceivers. Key fabrication parameters, such as repetition rate, laser power, scanning speed, and processing method, were optimized through parameter scanning. Using multi-scan and overwriting methods, we achieved low-loss waveguides with a propagation loss of 0.13 dB/cm and a coupling loss of 0.06 dB/facet at the wavelength of 1310 nm. To simultaneously suppress radiation loss and mode mismatch loss, we proposed a trench-assisted and offset-assisted cosine-shaped waveguide (TOA-cos) design. Simulations indicated that the TOA-cos design achieved a 1.0 dB/cm radiation-loss bending radius of only 2.5 mm at 1310 nm. Experimentally, the TOA-cos design achieved a low TE-mode insertion loss of 0.50 dB for compact S-bends ($1400 \times 190.5 \mu\text{m}^2$), compared to 3.0 dB for conventional arcs. Utilizing the TOA-cos design, a state-of-the-art FI/FO device was realized, featuring an insertion loss as low as 0.54 dB and a compact footprint as small as $2.5 \times 2.0 \text{ mm}^2$. Additionally, the proposed design is fully

compatible with existing FLDW bending-loss reduction methods, significantly enhancing its practical applicability for chip-level 3D optical interconnects in optical transceivers.

2 | Experimental Methods

The ultra-compact low-loss FI/FO device is fabricated using the FLDW system demonstrated in our previous work [9]. Linearly polarized femtosecond laser pulses (wavelength: 1030 nm, pulse duration: 370 fs, repetition rate: 300 kHz) are focused approximately $100 \mu\text{m}$ beneath the surface of boro-aluminosilicate glass substrates ($50 \times 10 \times 1.1 \text{ mm}^3$) using a $20\times$ dry objective (NA = 0.40). The power of the femtosecond laser is controlled by two linear polarizers and a half-waveplate. A three-axis precision stage with a positioning accuracy of $0.1 \mu\text{m}$ provides precise control of the focal point. The fabrication process is monitored using a charge-coupled device (CCD). Unless otherwise specified, all waveguides, including both straight and curved sections, have a total length of 10 mm.

Optical characterization of the insertion loss is performed at 1310 nm wavelength using a distributed feedback (DFB) laser source. The input light is coupled into the waveguide using a standard SMF (G.657.A1). The polarization of input light is controlled by a mechanical three-ring polarization controller. The output light is collected using an identical SMF and quantified with an optical power meter. To minimize Fresnel reflections at the interfaces, index-matching oil is applied to the waveguide end-facets. A back-to-back (B2B) power measurement is then performed by directly connecting the input and output fibers to assess losses induced by connectors. The final insertion loss values are determined by subtracting this B2B reference power from the measured output power through the waveguides.

3 | Principle of Reducing Bending Loss

The total insertion loss IL of bent waveguides comprises propagation loss L_p , coupling loss L_c , and bending loss L_b , which can be expressed as:

$$IL = L_p \times (l_s + l_b) + 2L_c + L_b \quad (1)$$

where l_s and l_b represent the length of straight waveguide and bent waveguide, respectively. Bending loss L_b can be further categorized into scattering loss L_s arising from sidewall roughness, radiation loss L_r along the curved segments, and mode mismatch loss L_{mismatch} occurring at the junctions. The scattering loss L_s refers to the excess loss of the bent waveguide over a straight one. This slight increase occurs because the mode field shifts closer to the sidewalls, causing stronger scattering. The bending loss of curved waveguides is expressed as follows:

$$L_b = L_r \times l_b + L_{\text{mismatch}} + L_s \times l_b \approx L_r \times l_b + L_{\text{mismatch}} \quad (2)$$

In simulations, scattering loss L_s is neglected. Radiation loss L_r and mode mismatch loss L_{mismatch} are governed by the geometric parameters of the waveguide curves. Radiation loss L_r scales with the curvature of bent waveguides. In S-bends, mode mismatch loss L_{mismatch} arises primarily at the interfaces between straight

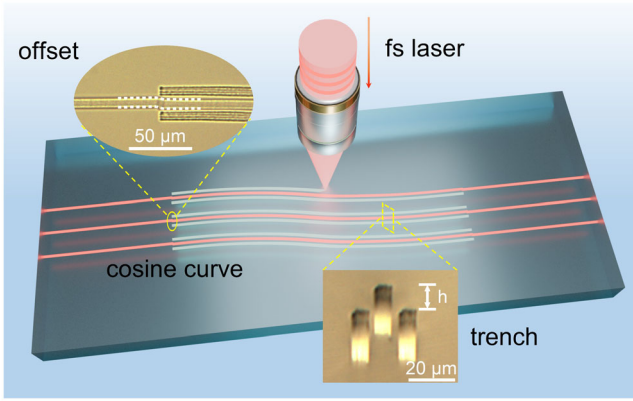


FIGURE 1 | Schematic of femtosecond laser direct-written TOA-cos waveguides.

and curved sections, as well as between adjacent curved segments, due to abrupt changes in the local curvature. While bending loss L_b differs for transverse electric (TE) and transverse magnetic (TM) polarizations, the numerical analysis is confined to the TE mode. Experimentally, however, both TE and TM modes are characterized to provide a comprehensive loss assessment.

Common methods for reducing bending loss in waveguides include increasing the refractive index contrast between the core and cladding [37–39], employing bends with adiabatically varying curvature [40–42], and introducing lateral offset at waveguide junctions [43–45]. Enhancing the refractive index difference improves mode confinement and reduces radiation loss L_r . Curves with adiabatic curvature can mitigate mode mismatch loss $L_{mismatch}$. The offset at the junction can improve mode field matching between adjacent waveguides with different curvature. Based on these three bending-loss-reduction methods, we propose a trench-assisted and offset-assisted cosine-shaped structure (TOA-cos), capable of simultaneously suppressing mode mismatch loss $L_{mismatch}$ and reducing radiation loss L_r , as shown in Figure 1.

The geometric shape and curvature profile of several curves are illustrated in Figure 2. Taking the arc-shaped bend (the green dotted curve) as an example, abrupt changes in curvature occur at points a and c between the straight waveguide (zero curvature) and the curved arc (constant non-zero curvature), as well as at point b between two opposing arcs (positive and negative curvature). These discontinuities result in mode mismatch, contributing to the overall bending loss. Furthermore, mode

mismatch loss $L_{mismatch}$ is also influenced by the rate of curvature change. Cosine and raised-sine [46] curves are commonly employed in the design of S-bends, as illustrated by the blue and orange solid lines in Figure 2. Although complex Euler [47] and Bézier [48] curves have also been demonstrated for S-shaped waveguides, it remains challenging to ensure both curvature continuity and low rates of curvature change. Thus, only cosine and raised-sine curves are considered here. Figure 2b shows the curvature of cosine and raised-sine curves. Although the curvature of the raised-sine curve reaches zero at points a and c, its higher rate of curvature change results in greater bending loss. Therefore, cosine curves are adopted in this work. The expressions of the cosine and raised-sine curves are as follows:

$$f(y) = \frac{\Delta x}{2} \times \left(1 - \cos \left(\pi \times \frac{y}{\Delta y} \right) \right) \quad (3)$$

$$f(y) = \Delta x \times \left(\frac{y}{\Delta y} - \frac{\sin(2\pi y / \Delta y)}{2\pi} \right) \quad (4)$$

where Δx denotes the center-to-center spacing between the input and output waveguides, Δy represents the horizontal routing length of the S-bend. The effective bending radius R_{eff} of the cosine-shaped waveguide is defined as the radius of an arc-shaped bend with identical Δx and Δy , as given by the following expression:

$$R_{eff} = \frac{(\Delta y^2 + \Delta x^2)}{4\Delta x} \quad (5)$$

Since the curvature of the cosine curve at points a and c is non-zero, a lateral offset is introduced to mitigate the mode mismatch, as illustrated in Figure 1. In curved waveguides, the optical mode field shifts toward the outer surface [49]. Introducing a lateral offset can enhance the overlap between adjacent mode fields, thereby reducing mode mismatch loss $L_{mismatch}$.

In order to enhance the refractive index contrast between waveguide and substrate, auxiliary trenches with a laser-induced refractive-index reduction are introduced on both sides of the guiding core, as illustrated in Figure 1. By fabricating these trenches with a height reduction h on both sides, the positive refractive index region of the guiding waveguides is sandwiched between the negative refractive index regions of the trenches. Unlike bend-loss-suppression walls [31], our approach relies

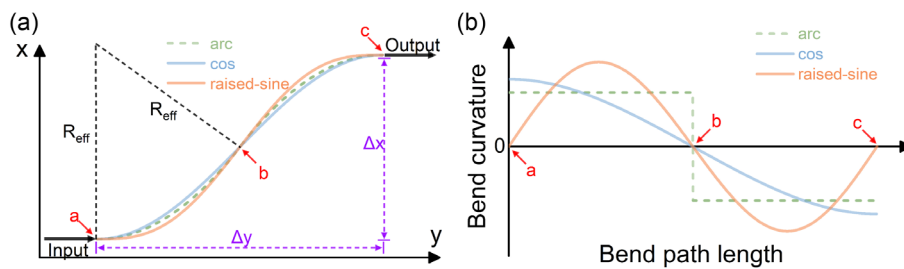


FIGURE 2 | (a) Schematic of S-bends formed by arc, cosine, and raised-sine curves. (b) Curvature as a function of the bending path length for each S-bend.

on the negative refractive-index contrast within the trenches, rather than stress-induced index modifications. The spacing between the two auxiliary trenches is fixed at $20\text{ }\mu\text{m}$ to ensure minimal impact on the guiding properties.

4 | Fabrication and Experimental Results

4.1 | Parameters of FLDW

The inscription parameters are first studied to achieve low-loss straight waveguides, including repetition rate, laser power, processing speed, and writing methods. Low-repetition-rate femtosecond lasers are employed for waveguide inscription, due to the ability to produce higher refractive index contrast compared to high-repetition-rate regimes [33]. A repetition rate of 300 kHz is finally determined, ensuring high processing efficiency and uniform waveguide quality. Owing to the minimal thermal accumulation associated with low-repetition-rate femtosecond laser processing, the fabricated waveguides exhibited highly confined cross-sectional dimensions. Thus, the multi-scan method has been employed [50]. Since repeatedly writing at the same location can increase the uniformity of waveguides, overwriting method has been employed. Based on the multi-scan and overwriting strategy, we initially overwrite m times at the starting position. The focal point is then shifted by a fixed displacement distance s , followed by an additional overwriting of m writing passes. This procedure is repeated sequentially in the same direction for n cycles, resulting in a total of $m \times n$ scanning passes throughout the entire fabrication process, as shown in Figure 3a. To ensure continuous adhesion between adjacent inscriptions and improve processing efficiency, the displacement distance s

is set to $1.0\text{ }\mu\text{m}$. The resulting writing methods are designated using the notation: $m \times n \times 1\text{ }\mu\text{m}$.

Utilizing multi-scan and overwriting methods, $2 \times 7 \times 1\text{ }\mu\text{m}$ straight waveguides are first fabricated for evaluating how insertion loss varies with scan speed and laser power. Our parameter optimization indicates optimal scan speed and laser power of $200\text{ }\mu\text{m/s}$ and 210 mW , respectively, as shown in Figure 3b,c. At the repetition rate of 300 kHz , the femtosecond laser delivers a pulse density of $1500\text{ pulses}/\mu\text{m}$ with a pulse energy of approximately $0.7\text{ }\mu\text{J/pulse}$. The resulting waveguide cross-sections with varying laser powers are demonstrated in Figure 3d. At low laser energy, only positive refractive index modification is induced. As the laser power increases, a characteristic waveguide structure forms, consisting of a bright region characterized by a positive refractive index change and a dark region exhibiting a negative refractive index variation compared to the glass substrate.

To reduce coupling loss with SMF and improve waveguide uniformity, various types of $m \times n \times 1\text{ }\mu\text{m}$ straight waveguides are fabricated and evaluated through parameter scanning, as illustrated in Figure 3e. When inscribing seven parallel tracks along the same direction ($n = 7$), the insertion loss of the waveguides reaches its minimum. Increasing the number of processing passes at the same position from $m = 2$ to $m = 4$ reduces the insertion loss by approximately 0.1 dB . Therefore, the $4 \times 7 \times 1\text{ }\mu\text{m}$ straight waveguides feature the insertion loss of 0.25 dB and the polarization-dependent loss (PDL) of less than 0.1 dB , with a measurement error of $\pm 0.05\text{ dB}$. However, in consideration of processing efficiency, configurations with higher values of m are not further investigated in loss measurement.

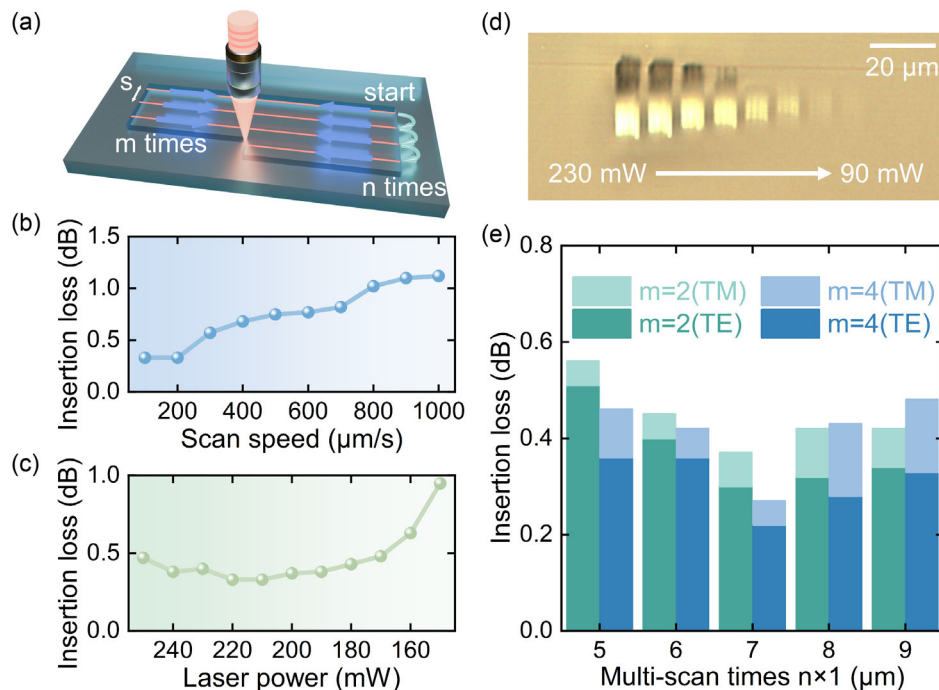


FIGURE 3 | (a) Schematic of multi-scan and overwriting methods. (b,c) Insertion losses of $2 \times 7 \times 1\text{ }\mu\text{m}$ straight waveguides as a function of scan speed at 210 mW and laser power at $200\text{ }\mu\text{m/s}$. (d) Cross-sectional views of waveguides fabricated at different laser power. (e) Insertion losses of various $m \times n \times 1\text{ }\mu\text{m}$ straight waveguides.

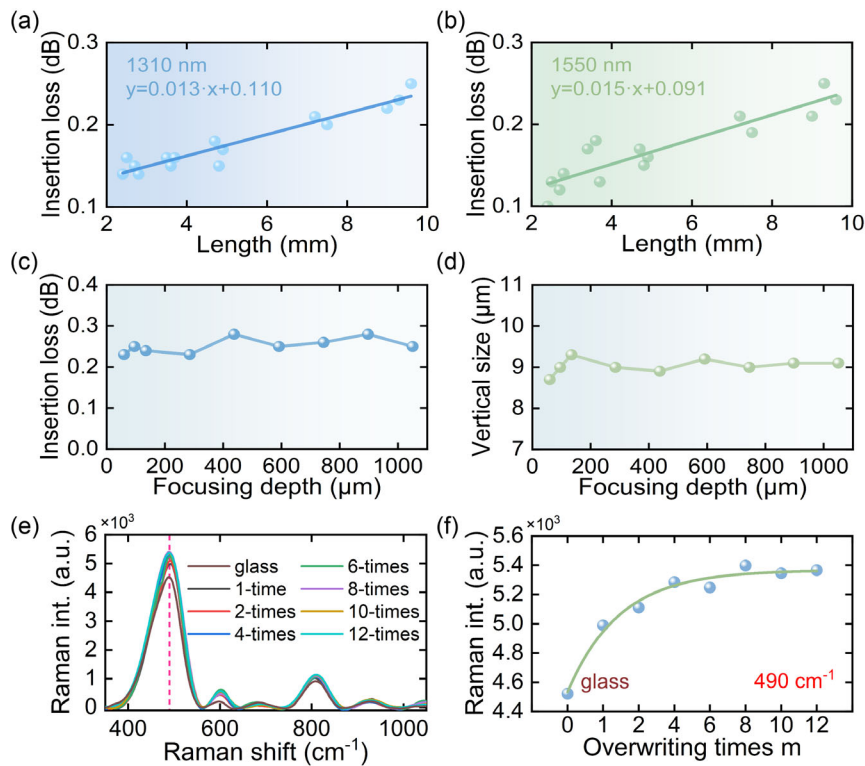


FIGURE 4 | (a,b) Cut-back methods of $4 \times 7 \times 1 \mu\text{m}$ straight waveguides at the wavelength of 1310 and 1550 nm. (c,d) Insertion losses and vertical dimensions of $4 \times 7 \times 1 \mu\text{m}$ straight waveguides with different inscription depth. (e) Raman spectra of waveguides with different overwriting times in the bright region. (f) Raman intensity at 490 cm^{-1} .

The propagation loss of about 0.13 dB/cm and the coupling loss of about 0.06 dB/facet are measured using cut-back method at the wavelength of 1310 nm, as shown in Figure 4a. The ultra-low coupling loss confirms excellent mode-field compatibility with SMF. Figure 4c,d present the insertion losses and vertical cross-sectional dimensions of waveguides written at different depths, demonstrating uniform fabrication capability across the 1.1 mm thick glass substrates. Spherical aberration is mitigated by the nonlinearity arising from laser power above the self-focusing threshold [51, 52]. The intensity-dependent positive lens counteracts aberration-induced defocusing, maintaining a high peak intensity at the writing depth and preventing focal elongation. Moreover, a moderate-NA objective ($\text{NA} = 0.40$) inherently reduces spherical aberration compared to a high-NA one, which enhances depth robustness. As a result, the vertical dimensions of waveguides exhibit minimal variation across different writing depths.

Raman spectroscopy is employed to analyze the influence of overwriting method on refractive index in the bright region, as shown in Figure 4e. The Raman intensity at 490 cm^{-1} increases with overwriting times m and eventually saturates. The enhanced Raman intensity at 490 cm^{-1} originates from the symmetrical stretching vibrations of bridging oxygens in four-membered rings. Since the conversion from higher-membered rings to four-membered rings corresponds to local densification of the glass, leading to higher refractive index [53], the refractive index in the bright region increases with overwriting times m . Consequently, overwriting is an effective technique for achieving higher refractive index, consistent with prior researches [36, 54]. Combining these Raman results with

the earlier loss measurements, the $4 \times 7 \times 1 \mu\text{m}$ waveguide configuration is confirmed as the optimal design for low-loss bent waveguides. Moreover, to avoid adding excessive fabrication time, the trenches are designed as $1 \times 7 \times 1 \mu\text{m}$ waveguides.

The refractive index distribution of the TOA-cos waveguide is shown in Figure 5. The characterization is performed using an advanced in situ 3D refractive-index imaging system [55]. A 633 nm He-Ne laser beam is split into reference and sample arms.

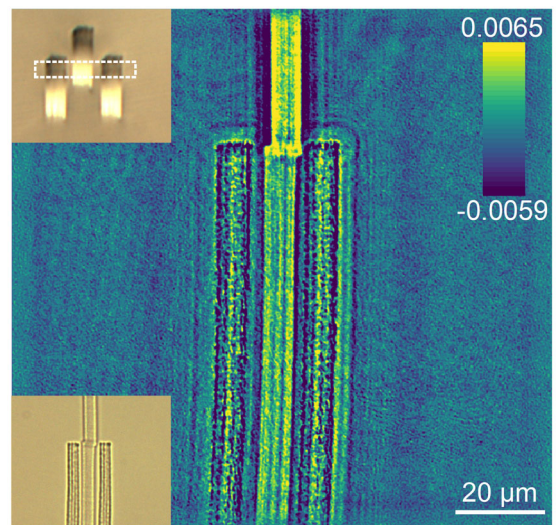


FIGURE 5 | The refractive index distribution of the TOA-cos waveguide. Top: cross-sectional view of the waveguide. White dotted lines mark the refractive index test positions. Bottom: top view of the waveguide.

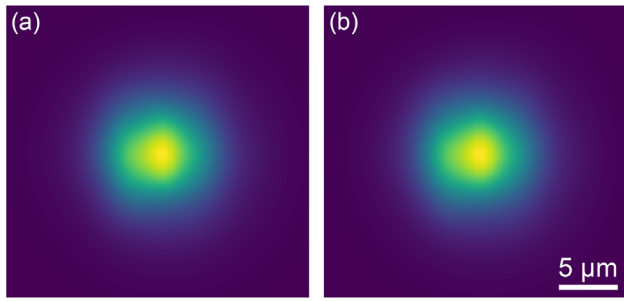


FIGURE 6 | Simulated mode fields of the (a) waveguide and (b) TA-waveguide at 1310 nm.

The sample beam is focused into the glass by a 20× objective and collected by a 50× oil-immersion objective on the opposite side. Then, it interferes with the reference beam on a CCD camera. The resulting interference fringes are recorded and analyzed to reconstruct the refractive-index distribution in regions modified by femtosecond laser. The measured refractive index differences are +0.0065 in the core and −0.0059 in the trench. The complex structures of waveguides may influence the measurements of refractive index, resulting in a nonuniform distribution in trenches.

4.2 | Simulated and Experimental Results of TOA-Cos Model

Simulations are performed to optimize the structure of TOA-cos waveguides according to the measured refractive index distribution using the beam propagation method (BPM). To study

trench-induced mode field changes, we simulated the mode fields of standard and trench-assisted waveguides, as shown in Figure 6. The overlap integral of the two modes is 99.9%, indicating that the mode mismatch is negligible between standard and trench-assisted waveguides.

The bending losses of curved waveguides with various footprints are achieved at the output ports. The total insertion losses of bent waveguides are calculated by summing the bending loss with the insertion loss of a straight waveguide of equivalent optical path length. Radiation loss is quantified by dividing the bending loss by the propagation length of the curved section, excluding transitional regions where light intensity fluctuates due to mode mismatch effects.

Different trench depth h is simulated to minimize insertion loss, as presented in Figure 7a. The simulation results reveal that the insertion loss is minimized at a trench depth h of 10.0 μm , which exhibits no dependence on the bending radius. At larger effective bending radius R_{eff} , the insertion loss exhibits a plateau when the trench depth h exceeds 10.0 μm , indicating that the trench design becomes effective for R_{eff} below 3.8 mm. Additionally, for trench depth h below 10.0 μm , evanescent coupling occurs between guiding waveguide and the bright regions of trenches, leading to higher insertion loss. The influence of offset across various bending radii is also simulated, as illustrated in Figure 7b. The simulations reveal that employing an optimal lateral offset significantly minimizes insertion loss, and the optimal value is dependent on the bending radius. Thanks to the high positioning accuracy (0.1 μm) of FLDW system, such precise offset can be

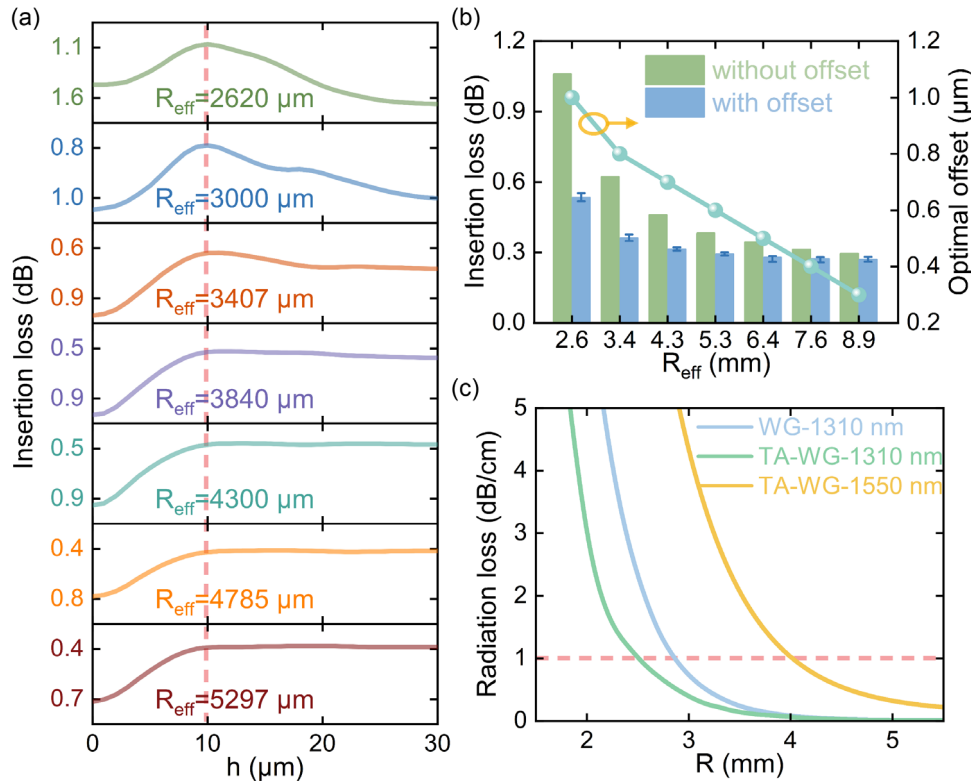


FIGURE 7 | (a) The simulated insertion loss of TA-cos waveguides under different trench depth. (b) The simulated insertion loss of TA-cos waveguides with and without optimal lateral offset. (c) The simulated radiation loss of waveguides with and without trench assistance. The value of Δx in (a) and (b) is fixed at 190.5 μm .

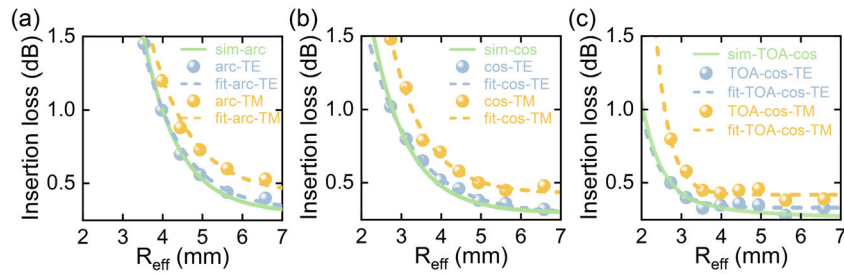


FIGURE 8 | Measured insertion losses for TE and TM modes of (a) arc-shaped S-bends, (b) cosine-shaped S-bends, (c) TOA-cos S-bends. The value of Δx is fixed at 190.5 μm .

reproducibly fabricated. Furthermore, the error bar in Figure 7b represents the insertion loss fluctuation caused by the fabricated offset deviation of $\pm 0.2 \mu\text{m}$. The insertion loss exhibits negligible sensitivity to minor variations in lateral offset during fabrication, demonstrating the robustness of the proposed TOA-cos design against processing tolerances. To more accurately evaluate the bending loss L_b , we quantitatively assessed the radiation loss L_r at various bending radii, as shown in Figure 7c. The radiation loss L_r of the TA-waveguide is lower than that of waveguide without trenches. Furthermore, the 1.0 dB/cm cutoff bending radius reaches 2.5 mm at a wavelength of 1310 and 4.0 mm at a wavelength of 1550 nm.

The measured insertion losses of different bent waveguides are presented in Figure 8. The S-bend waveguides are fabricated with a fixed Δx of 190.5 μm . Insertion losses are characterized for both TE and TM modes at the wavelength of 1310 nm. Figure 8a–c present the insertion losses of arc-shaped S-bends, cosine-shaped S-bends, and TOA-cos S-bends, respectively. Under the configuration of Δy of 1400 μm , Δx of 190.5 μm , and total length of 10 mm, the TOA-cos waveguide achieves a TE-mode insertion loss of 0.5 dB. This performance significantly surpasses that of the cosine-shaped S-bend (1.0 dB) and the arc-shaped S-bend (3.0 dB) under identical parameters. The waveguide has an effective bending radius of 2.62 mm, and the simulated radiation loss is 0.85 dB/cm. The ideal insertion loss is obtained by adding the coupling loss ($0.06 \text{ dB} \times 2 = 0.12 \text{ dB}$), propagation loss ($0.13 \text{ dB/cm} \times 1.0 \text{ cm} = 0.13 \text{ dB}$), and radiation loss ($0.85 \text{ dB/cm} \times 1.4 \text{ cm} = 0.12 \text{ dB}$), which sums to 0.37 dB. Consequently, the mode-mismatch loss is calculated to be 0.13 dB. The measured insertion losses for the TE mode align well with the simulated values, which confirms the validity of the simulation model and its capability to accurately predict the loss characteristics of bending waveguides. Furthermore, as the effective bending radius R_{eff} decreases, the bending loss of TM mode becomes more pronounced, indicating an increase in PDL with reduced bending radius. This is attributed to the TM mode's weaker modal confinement, which makes it more susceptible to leaking into radiation modes in bent waveguides. Consequently, the TM mode leaks more strongly than the TE mode, causing greater polarization dependence in tighter bends.

Utilizing the TOA-cos model, a compact four-channel FI/FO device and several bent waveguides with large footprints are demonstrated. As illustrated in Figure 9a, the ultra-compact FI/FO device is designed to interface a four-core fiber with a 40 μm core pitch and a waveguide array with a channel separation of 127 μm inside optical transceiver. Each channel comprises

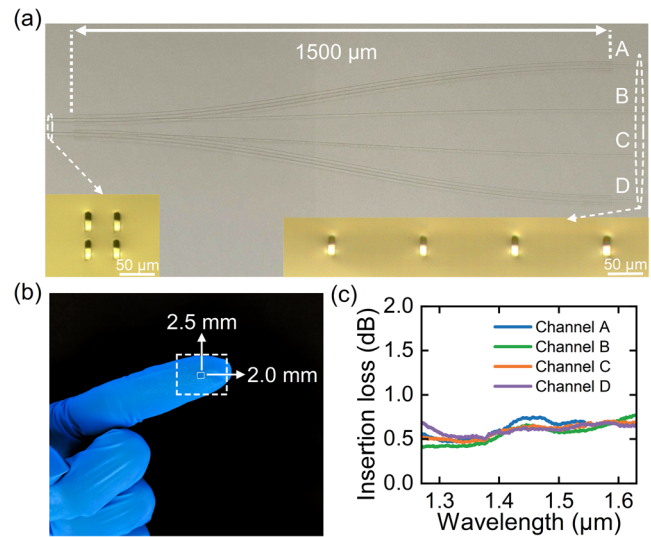


FIGURE 9 | (a) Micrograph and (b) photograph of the ultra-compact 3D FI/FO device. (c) The spectra of four channels of FI/FO device.

a 1.5 mm curved waveguide with 0.5 mm straight waveguides on both ends. The device achieves a minimal effective bending radius R_{eff} of 3.34 mm within a total footprint of only $2.5 \times 2.0 \text{ mm}^2$. The TOA-cos waveguides are employed only for small effective bending radii R_{eff} of 3.34 mm, whereas cosine-shaped S-bends are utilized for larger radii. Figure 9c demonstrates the spectra of the four channels, indicating that the FI/FO device is suitable for all telecommunication wavelength. The insertion losses for all four channels are less than 0.54 dB at the wavelength of 1310 nm. To the best of our knowledge, this FI/FO device exhibits the smallest footprint reported to date using FLDW in glass substrates, representing a state-of-the-art achievement in miniaturization. Although this FI/FO device has not been fully integrated with MCFs and photonic chips, our earlier work has demonstrated that assembly-induced loss would be acceptable for practical application [9]. In addition, multiple S-bend waveguides with large center-to-center spacing Δx are successfully fabricated, as presented in Figure 10. The TOA-cos curved waveguide, with dimensions of $\Delta x = 8.0 \text{ mm}$ and $\Delta y = 4.0 \text{ mm}$, achieves a low TE mode insertion loss of 0.60 dB and a PDL of 0.15 dB. The experimentally measured insertion loss is slightly higher than the simulated value, which can be attributed to minor imperfections in the fabrication process. The excellent performance of large-angle bent waveguides verifies the multi-channel inscribing capability of the proposed model in compact, high-density 3D-PIC.

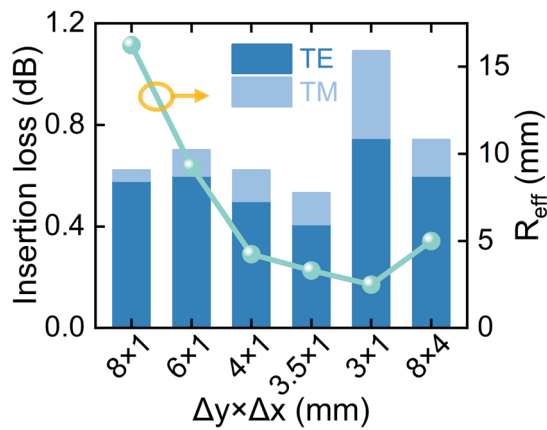


FIGURE 10 | Insertion losses of curved waveguides with large footprints.

5 | Discussion and Conclusion

The TOA-cos curved waveguide model demonstrates a marked advantage over conventional arc-shaped S-bends. Figure 11 presents a systematic comparison of the simulated insertion losses for arc-shaped and TOA-cos S-bends across various Δx and Δy values. The white dotted lines represent the insertion loss threshold of 1.0 dB. The results clearly demonstrate the superior performance of the TOA-cos model under identical waveguide parameters. In practical fabrication, however, fabrication imperfections may lead to higher experimental losses compared to theoretical predictions. Moreover, the footprint of the TOA-cos waveguides remains a primary consideration for high-density integration. The longitudinal dimension could potentially be reduced by optimizing the trench geometry, such as the application of lens with high numerical aperture (NA). In contrast to prior research focusing on mitigating radiation loss L_r , this work introduces a novel method that simultaneously suppresses mode mismatch and enhances the refractive-index contrast in curved waveguides. Consequently, the TOA-cos design can be integrated with existing bending-loss-reduction approaches for FLDW, such as stress engineering [31] or composite waveguide structures [33]. This excellent compatibility significantly broadens the practical applicability of our method within the FLDW-based compact 3D-PICs.

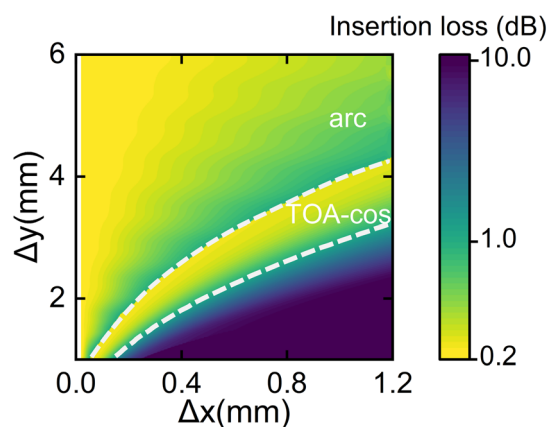


FIGURE 11 | Simulated insertion losses of arc-shaped and TOA-cos S-bends across varying footprints.

In summary, we demonstrated an ultra-compact, low-loss FI/FO device based on FLDW for interconnection between MCF and high-speed optical transceiver. The fabrication parameters were systematically optimized, including repetition rate, laser power, scan speed, and processing methodology. Employing multi-scan and overwriting methods, the propagation loss and coupling loss of the $4 \times 7 \times 1 \mu\text{m}$ waveguides were optimized to 0.13 dB/cm and 0.06 dB/facet at the wavelength of 1310 nm, using a repetition rate of 300 kHz, pulse energy of 0.7- μJ /pulse and scan speed of 200 $\mu\text{m/s}$. Different sources of bending loss were investigated, with corresponding optimization methods developed. Auxiliary trenches were used to reduce radiation loss, while cosine curves combined with lateral offsets were employed to mitigate mode mismatch loss. Both simulated and experimental approaches were utilized to validate the effectiveness of the proposed TOA-cos model. Simulations indicated that the bending radius of 1.0 dB/cm radiation loss was as low as 2.5 mm at 1310 nm. Experimentally, the TOA-cos design reduced the TE-mode insertion loss to 0.50 dB for compact S-bends ($1400 \times 190.5 \mu\text{m}^2$), representing an 83% reduction compared to conventional arcs (3.0 dB). Utilizing the TOA-cos design, we successfully demonstrated a state-of-the-art FI/FO device featuring a low insertion loss as small as 0.54 dB combined with an exceptionally compact footprint as small as $2.5 \times 2.0 \text{ mm}^2$. To the best of our knowledge, this low-loss FI/FO device represents the most compact FLDW-based design in glass substrates reported to date, which acts as a critical interface to interconnect MCFs with high-speed optical transceivers. Our method enables superior performance in high-density, chip-level optical interconnects while maintaining full compatibility with existing FLDW-based bending-loss-reduction methods, positioning it as a versatile solution for ultra-compact 3D photonic circuits.

Author Contributions

Fengrui Yu: conceptualization, methodology, validation, investigation, writing – review and editing, writing – original draft, formal analysis, data curation. **Lin Ma:** conceptualization, methodology, resources, project administration, supervision, writing – review and editing, funding acquisition. **Mingjing Xu:** writing – review and editing. **Jinhua Wu:** conceptualization, methodology. **Zuyuan He:** resources, project administration, writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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